=== UNIT 1: CELL ARCHITECTURE, MEMBRANE DYNAMICS & TRANSPORT MECHANISMS ===

>> 1.1 Structural Foundations: Prokaryotic vs. Eukaryotic Cells

Cell biology forms the foundational paradigm of biotechnology. Living organisms are broadly categorized based on their ultrastructural organization into Prokaryotes (domains Bacteria and Archaea) and Eukaryotes (animals, plants, fungi, and protists).
* Prokaryotic Architecture: Prokaryotes lack membrane-delimited intracellular organelles. Their genetic material consists of a single, covalently closed circular double-stranded DNA molecule situated in an irregularly shaped cytoplasmic region known as the nucleoid. They possess 70S ribosomes (composed of a 50S large subunit and a 30S small subunit). The cell envelope comprises the plasma membrane, a peptidoglycan cell wall (thick in Gram-positive bacteria, thin with an outer lipopolysaccharide membrane in Gram-negative bacteria), and occasionally an outer glycocalyx capsule.
* Eukaryotic Compartmentalization: Eukaryotes possess a defined double-membrane nucleus containing linear chromosomes compacted with basic histone proteins into chromatin fibers. Compartmentalization enables distinct biochemical reactions to proceed concurrently under tailored microenvironmental conditions (e.g., acidic pH in lysosomes vs. alkaline pH in the mitochondrial matrix). Cytoplasm contains 80S ribosomes (60S and 40S subunits) and an extensive endomembrane system.

>> 1.2 Fluid Mosaic Model of Biological Membranes

Proposed by S.J. Singer and Garth L. Nicolson in 1972, the Fluid Mosaic Model remains the universal foundation for membrane structure.
* Lipid Bilayer Matrix: Amphipathic phospholipids spontaneously assemble into a continuous bilayer in aqueous media. Polar hydrophilic headgroups (glycerol backbone esterified to phosphate and choline/ethanolamine) orient toward the intra- and extracellular fluids, while hydrophobic nonpolar fatty acid tails (saturated and cis-unsaturated hydrocarbon chains) pack into the hydrophobic core.
* Membrane Fluidity Regulators: Fluidity is determined by fatty acid saturation and temperature. Cis-double bonds create 30-degree kinks preventing dense crystalline packing, lowering the phase-transition temperature (Tm). In animal cells, cholesterol acts as a bidirectional fluidity buffer: at physiological temperatures (37 deg C), its rigid steroid planar rings restrain phospholipid lateral movement, reducing permeability; at low temperatures, it prevents close hydrocarbon tail packing, preventing membrane crystallization.
* Membrane Asymmetry: The inner (cytosolic) and outer (exoplasmic) leaflets exhibit distinct lipid compositions. Phosphatidylcholine and sphingomyelin predominate in the outer leaflet, whereas phosphatidylethanolamine and negatively charged phosphatidylserine are concentrated in the inner leaflet. Flip-flop (transverse) movement between leaflets is thermodynamically unfavored and catalyzed by

ATP-dependent enzymes: Flippases, Floppases, and Scramblases. External exposure of phosphatidylserine serves as a universal eat-me signal for apoptotic phagocytosis.

>> 1.3 Membrane Proteins and Glycocalyx

* Integral (Transmembrane) Proteins: Span the hydrophobic interior completely, typically utilizing alpha-helical bundles (e.g., bacteriorhodopsin, GPCRs with 7 transmembrane helices) or beta-barrels (e.g., porins in bacterial and mitochondrial outer membranes).
* Peripheral Proteins: Attached non-covalently via electrostatic interactions and hydrogen bonding to membrane surfaces or integral proteins. Readily extracted by high ionic strength buffers or pH shifts.
* Lipid-Anchored Proteins: Covalently attached to acyl chains (myristoylation, palmitoylation) or glycosylphosphatidylinositol (GPI) anchors.

>> 1.4 Transport Across Biological Membranes

* Simple Diffusion: Passive flux of small, non-polar molecules (O2, CO2, N2, steroid hormones) driven exclusively by concentration gradients according to Fick's First Law: J = -P * (C2 - C1).
* Facilitated Diffusion: Protein-mediated passive transport exhibiting saturation kinetics following Michaelis-Menten principles. Utilizes carrier proteins (GLUT uniporters undergoing conformational shifts) and ion channels (voltage-gated, ligand-gated, mechanosensitive) which transport ions at rates approaching 10^7 ions/sec.
* Primary Active Transport: Directly couples ATP hydrolysis to solute translocation against electrochemical gradients. Exemplified by the Na+/K+ ATPase, an electrogenic P-type pump that hydrolyzes 1 ATP to export 3 Na+ ions and import 2 K+ ions, establishing cellular resting membrane potential (-70 mV) and maintaining cell volume.
* Secondary Active Transport: Utilizes the potential energy stored in an electrochemical gradient created by primary pumps to drive transport of another solute against its gradient. Symport (co-transport) moves both solutes in the same direction (e.g., SGLT1 Na+/glucose transporter in intestinal enterocytes). Antiport (counter-transport) moves solutes in opposite directions (e.g., Na+/H+ exchanger, Na+/Ca2+ exchanger).
* Bulk Transport: Endocytosis (cellular uptake via vesicular budding) subdivided into Phagocytosis ('cell eating' of large particulates by specialized macrophages), Pinocytosis ('cell drinking' of fluid phase solutes), and Receptor-Mediated Endocytosis (clathrin-coated pits capturing LDL via LDL receptors). Exocytosis mediates targeted secretory vesicle fusion via SNARE proteins.

[EXAMPLE 1: The Cardiac Glycoside Mechanism on Na+/K+ ATPase]

Problem Context: Digitalis and ouabain are plant-derived cardiotonic steroids used clinically in congestive heart failure. Explain the molecular cascade and calculate why inhibiting the primary active Na+/K+ pump enhances cardiac muscle contraction velocity.
Step 1 (Pump Inhibition): Digitalis binds stereospecifically to the extracellular pocket of the phosphorylated E2-P conformation of Na+/K+ ATPase, halting ATP-dependent dephosphorylation and pump recycling.
Step 2 (Intracellular Sodium Elevation): Because primary Na+ extrusion is diminished while Na+ leak channels remain functional, cytosolic [Na+] rises from baseline ~10 mM to ~25 mM, dissipating the steep transmembrane Na+ concentration gradient.
Step 3 (Secondary Active Transporter Suppression): The cardiac sarcolemmal 3Na+/1Ca2+ antiporter (NCX) relies on the free energy of downhill Na+ influx to drive uphill Ca2+ extrusion. The reduced chemical potential gradient of Na+ dramatically reduces the driving force for Ca2+ efflux or causes reverse mode operation.
Step 4 (Sarcoplasmic Reticulum Loading): Elevated cytosolic [Ca2+] is sequestered by the SERCA2a pump into the sarcoplasmic reticulum. Upon subsequent electrical depolarization, an amplified burst of Ca2+ is released through Ryanodine receptors into the myoplasm.
Conclusion: Ca2+ binds Troponin C with higher saturation, promoting stronger actin-myosin cross-bridge cycling and inducing positive inotropic myocardial contractility.

[EXAMPLE 2: Michaelis-Menten Kinetics of Glucose Transporter (GLUT1)]

Problem Context: Erythrocyte glucose uptake is mediated by the facilitated uniporter GLUT1. If Km for D-glucose is 1.5 mM and Vmax is 30 micromol/min/10^8 cells, calculate the transport flux (V) at physiological fasting blood glucose (5.0 mM) and severe hypoglycemic condition (1.0 mM). Compare with D-galactose (Km = 30 mM).
Formula: Michaelis-Menten hyperbolic saturation relation: V = (Vmax * [S]) / (Km + [S]).
Calculation 1 (Physiological Fasting Glucose, [S] = 5.0 mM):
V_fasting = (30 * 5.0) / (1.5 + 5.0) = 150 / 6.5 = 23.08 micromol/min/10^8 cells.
Operating saturation percentage = 23.08 / 30 = 76.9% of maximal velocity.
Calculation 2 (Severe Hypoglycemia, [S] = 1.0 mM):
V_hypoglycemia = (30 * 1.0) / (1.5 + 1.0) = 30 / 2.5 = 12.00 micromol/min/10^8 cells.
Operating saturation drops to 40% of Vmax, explaining clinical neurological symptoms (neuroglycopenia) due to glucose depletion in brain erythrocytes.
Calculation 3 (Substrate Specificity for D-galactose, [S] = 5.0 mM, Km = 30 mM):
V_galactose = (30 * 5.0) / (30 + 5.0) = 150 / 35 = 4.29 micromol/min/10^8 cells.
Conclusion: GLUT1 shows a 5.4-fold higher transport rate for D-glucose over D-galactose at identical substrate concentrations due to lower Km (higher binding affinity).

[EXAMPLE 3: Receptor-Mediated Endocytosis and Familial Hypercholesterolemia]

Problem Context: Characterize the endocytic pathway of Low-Density Lipoprotein (LDL) and explain the biochemical defects leading to severe atherosclerosis in Familial Hypercholesterolemia (FH).

Pathway Details:

1. Circulating LDL particles contain a central hydrophobic core of 1,500 cholesteryl esters enclosed by an amphipathic monolayer of phospholipids and a single massive apolipoprotein B-100 (ApoB-100).
2. ApoB-100 binds to high-affinity LDL receptors clustered in clathrin-coated pits on the plasma membrane. Adaptor protein AP2 links the cytoplasmic NPXY internalization signal motif of the LDL receptor to the clathrin triskelion lattice.
3. Dynamin, a large GTPase, forms a helical collar around the neck of the invaginating coated pit and hydrolyzes GTP to pinch off the clathrin-coated endocytic vesicle.
4. Vesicles shed their clathrin coat via Hsp70 uncoating ATPase and fuse with early endosomes. The acidic lumen of early endosomes (pH 6.0 maintained by V-type H+ ATPase) induces conformational release of LDL from its receptor.
5. The LDL receptor is recycled via transport vesicles back to the plasma membrane (cycling ~100 times in its 20-hour lifespan). LDL particles are delivered to lysosomes (pH 4.8), where acid lipases degrade cholesteryl esters into unesterified free cholesterol.

Clinical Case Application (FH Mutation Classes): Class 1: No receptor synthesis; Class 2: Defective transport from ER to Golgi; Class 3: Defective LDL binding domain; Class 4: Mutations in the cytoplasmic NPXY sequence preventing pit localization. In all cases, lack of clearance leads to plasma cholesterol levels >600 mg/dL and juvenile coronary artery disease.

[EXAMPLE 4: Energetics of Secondary Active Glucose Transport (SGLT1)]

Problem Context: In the brush border membrane of intestinal epithelial cells, the SGLT1 cotransporter moves 1 molecule of D-glucose into the cell coupled to the influx of 2 Na+ ions. Derive the thermodynamic equation and determine whether an extracellular [Na+] of 140 mM, intracellular [Na+] of 14 mM, and membrane potential of -70 mV can accumulate glucose inside the cell against a 1,000-fold concentration gradient at 37 deg C (310 K).

Thermodynamic Relation: For 2 Na+ coupled transport, free energy change delta G = delta G_glucose + 2 * delta G_Na.

At equilibrium, delta G = 0: delta G_glucose = -2 * delta G_Na.

delta G_glucose = R * T * ln([Glucose_in] / [Glucose_out])

delta G_Na = R * T * ln([Na_in] / [Na_out]) + z * F * Vm

Values: R = 8.314 J/mol*K, T = 310 K, F = 96485 C/mol, Vm = -0.070 V, [Na_in]/[Na_out] = 14/140 = 0.1, z = +1.

Calculation of Electrochemical Energy per mol Na+:

delta G_Na = (8.314 * 310 * ln(0.1)) + (1 * 96485 * (-0.070))

delta G_Na = (2577.34 * (-2.3026)) - 6753.95 = -5934.57 - 6753.95 = -12,688.5 J/mol = -12.69 kJ/mol.

Available Driving Energy for 2 Na+ ions = 2 * (-12.69 kJ/mol) = -25.38 kJ/mol.
Required Energy for 1,000-fold Glucose Gradient accumulation:
delta G_req = R * T * ln(1000) = 2577.34 * 6.9078 = +17,803.7 J/mol = +17.80 kJ/mol.
Thermodynamic Evaluation: Since available coupling energy |-25.38 kJ/mol| exceeds the required energy |+17.80 kJ/mol|, the process is exergonic (delta G_total = +17.80 - 25.38 = -7.58 kJ/mol < 0).
Conclusion: SGLT1 easily drives a 1,000-fold uphill accumulation of glucose from the intestinal lumen into the enterocyte cytoplasm.

>> 1.5 University Review & Examination Questions

* Q1: Contrast the structural organization of Gram-positive vs. Gram-negative bacterial envelopes and explain why Gram-negative bacteria exhibit higher resistance to beta-lactam antibiotics.
Answer: Gram-positive cell walls have a thick (20-80 nm) multi-layered peptidoglycan meshwork laced with teichoic and lipoteichoic acids, directly exposed to the external medium. Gram-negative envelopes feature a thin (2-7 nm) single-layered peptidoglycan sheet within a periplasmic space bounded by an asymmetrical outer membrane. The outer leaflet contains lipopolysaccharide (Lipid A endotoxin, core oligosaccharide, O-antigen). The outer membrane acts as an impermeable steric and hydrophobic barrier; beta-lactams must diffuse through narrow porin channels. Gram-negative species frequently mutate porins or express periplasmic beta-lactamases, rendering them significantly more resistant.
* Q2: Explain the structural and thermodynamic factors governing membrane lipid asymmetry.
Answer: Biological membranes feature asymmetric distribution: outer leaflet is rich in phosphatidylcholine and sphingomyelin; inner leaflet is rich in phosphatidylethanolamine, phosphatidylinositol, and negatively charged phosphatidylserine. Spontaneous transverse flip-flop is energetically unfavorable because polar headgroups must traverse the central 3-4 nm hydrophobic core. Asymmetry is maintained by ATP-driven Flippases (P4-ATPases that translocate PE/PS inward), Floppases (ABC transporters that pump PC/cholesterol outward), and Scramblases (Ca2+-activated bidirectional non-specific scramblers active during platelet coagulation and apoptosis).

>> 1.6 Key Takeaways and Revision Points

* The Singer-Nicolson Fluid Mosaic model describes membranes as 2D liquid crystals where lipids and proteins diffuse laterally.
* Cholesterol acts as a bidirectional membrane fluidity buffer, restraining movement at high temperatures and preventing solid freezing at low temperatures.
* Primary active transport utilizes ATP directly (P-type, V-type, F-type, ABC), while secondary active transport harnesses ion gradients generated by primary pumps.
* Clathrin, AP2, dynamin, and rab GTPases orchestrate targeted endocytic vesicle formation, cargo sorting, and lysosomal degradation.